

GTIIT Spring 2025 Calculus 1M

Workshop 11

May 20, 2025

Try to solve the following exercises. Feel free to chat about them with your classmates, but make sure to write your own solutions. If you get stuck, just move on to the next one and come back later.

If you need any help at all —whether it's a tiny hint or a big clue— don't hesitate to ask me!

Important note: If you find any errors or typos, please let me know so I can correct them.

Exercise 1. Determine whether the following integrals converge or diverge. If they converge, find their value.

(i) $\int_0^\infty \frac{1}{\sqrt[4]{1+x}} dx$

(ii) $\int_{-2}^{14} \frac{1}{\sqrt[4]{x+2}} dx$

(iii) $\int_0^{\pi/2} \frac{\cos \theta}{\sqrt{\sin \theta}} d\theta$

Exercise 2. (i) Evaluate the integral $\int_0^\infty x^n e^{-x}$ for $n = 0, 1, 2$ and 3.

(ii) Guess the value of $\int_0^\infty x^n e^{-x} dx$ when n is a positive integer.

(iii) Prove your guess by using mathematical induction.

Exercise 3. Find the length of the curves:

(i) $y = 2 \ln(\sin \frac{x}{2}), \pi/2 \leq x \leq \pi$

(ii) $12x = 4y^3 + 3y^{-1}, 1 \leq y \leq 3$

Exercise 4. Find the volume obtained when the circle of radius 1 with center $(1, 0)$ is rotated about the y -axis.